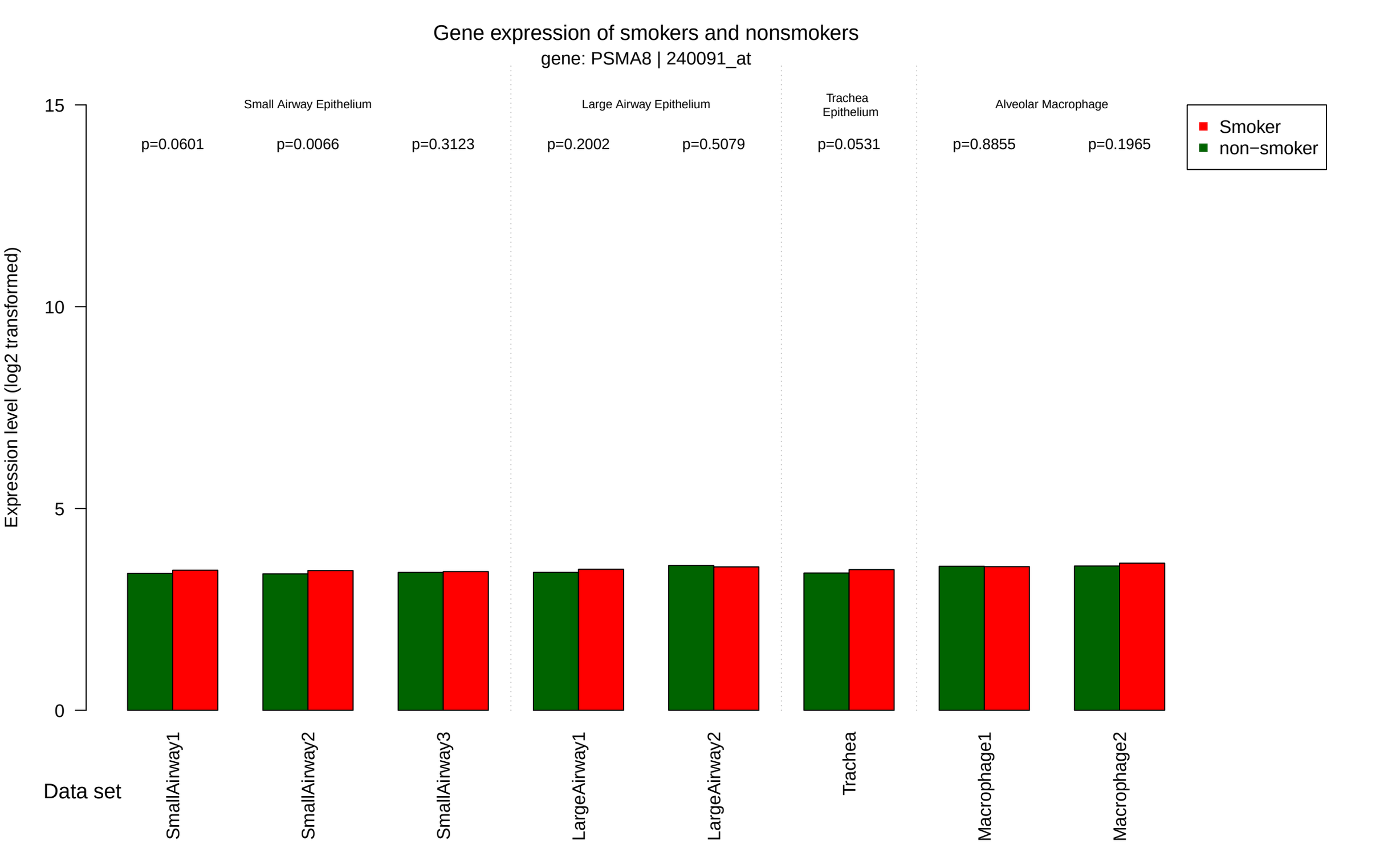
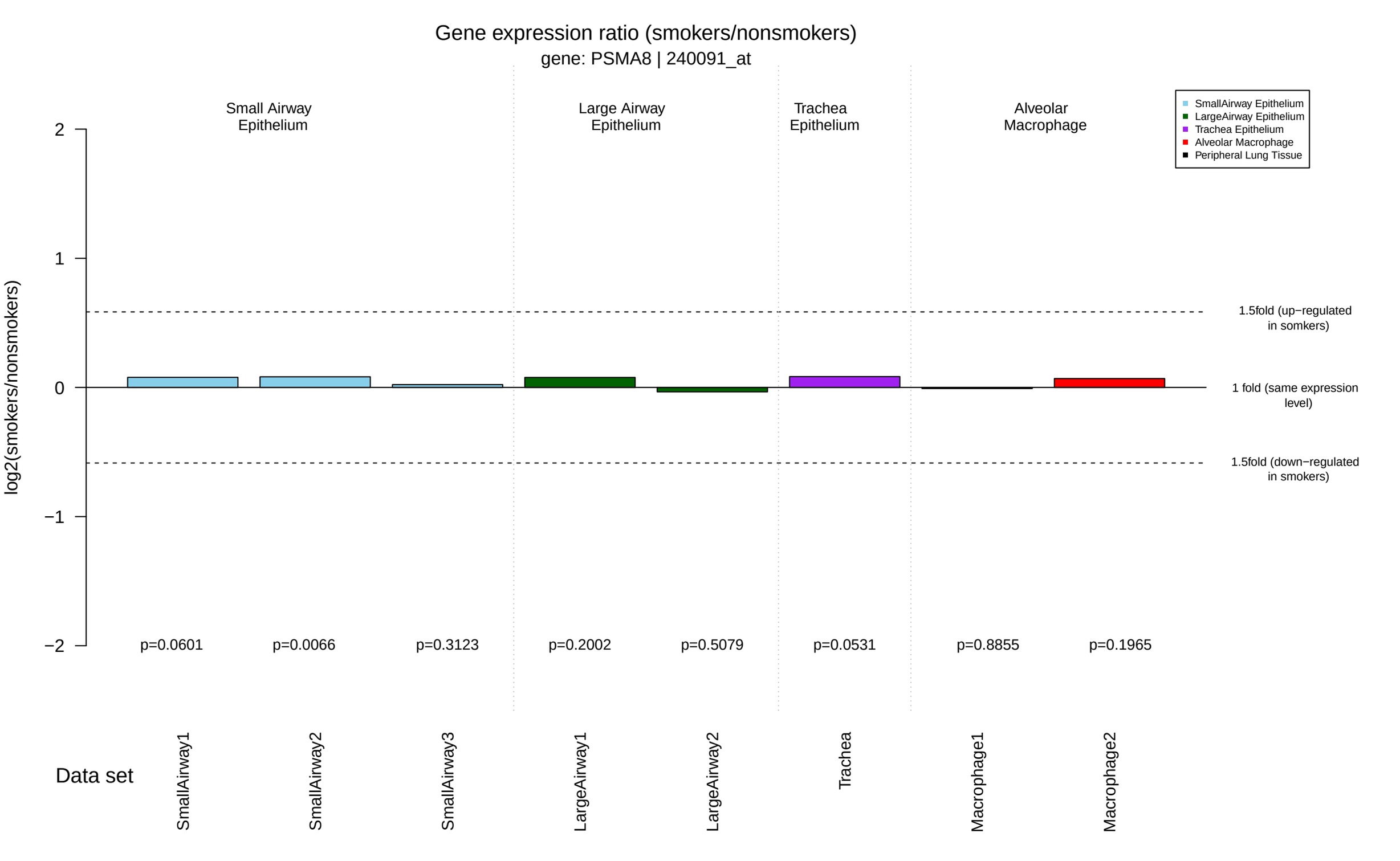
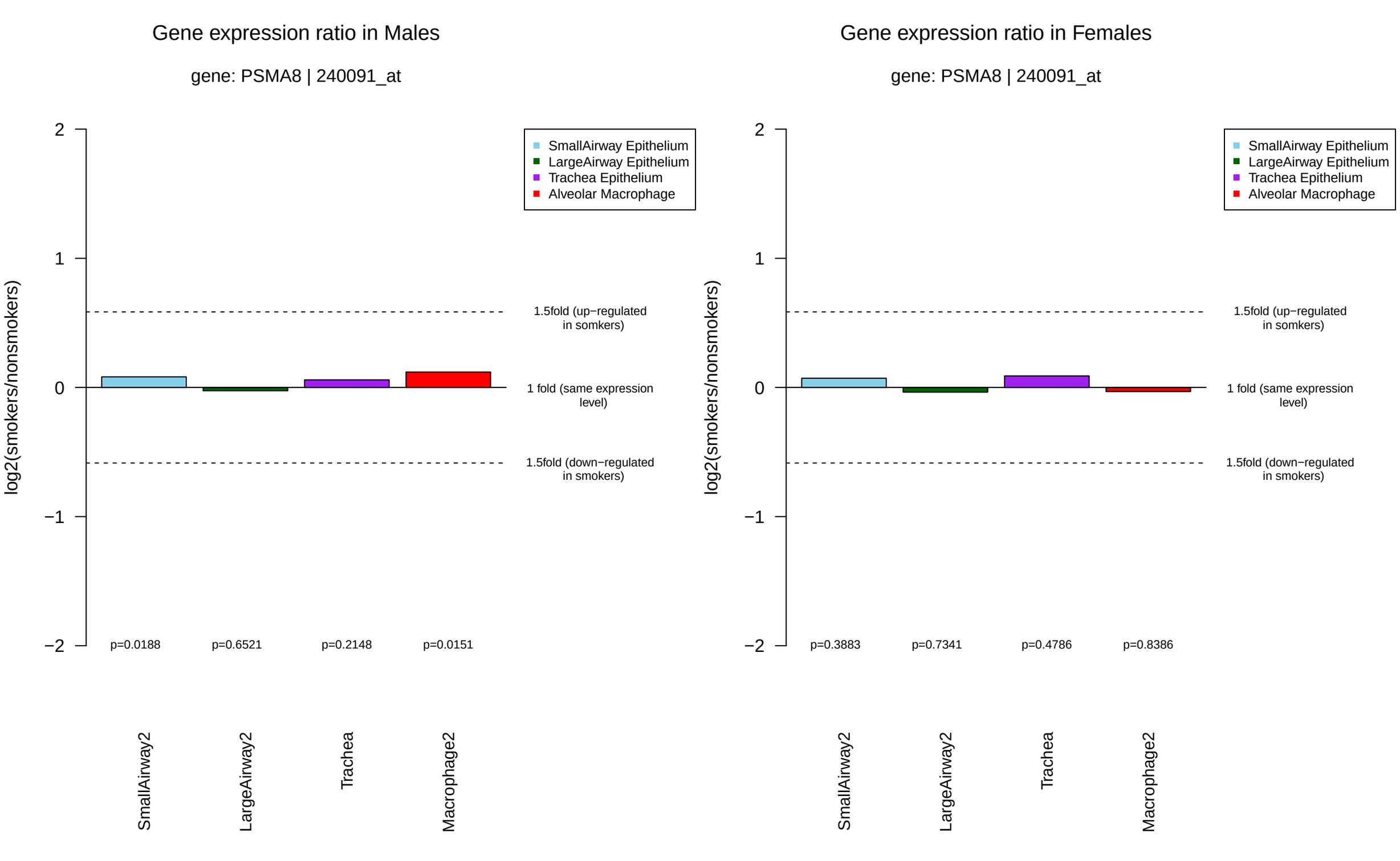
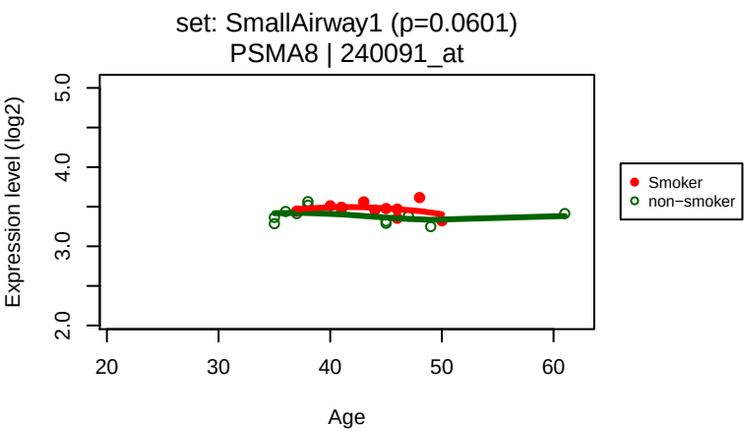
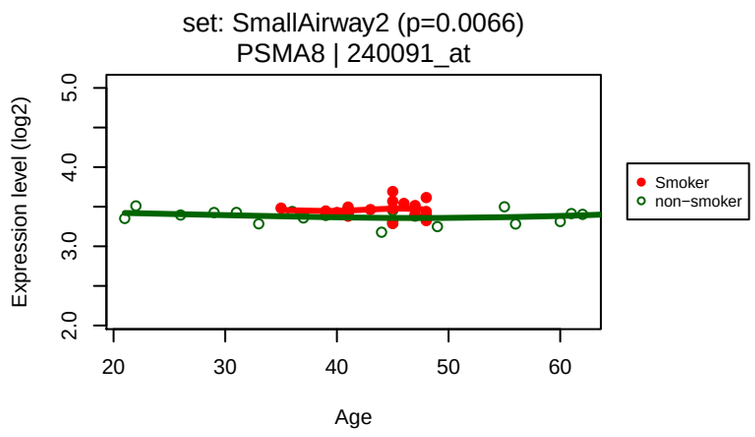
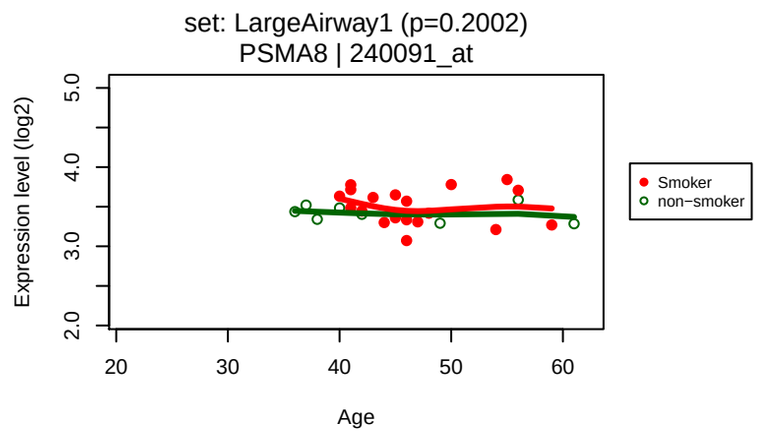

Gene expression of smokers and nonsmokers
gene: PSMA8 | 240091_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0601
p=0.0066
p=0.3123
p=0.2002
p=0.5079
p=0.0531
p=0.8855
p=0.1965
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

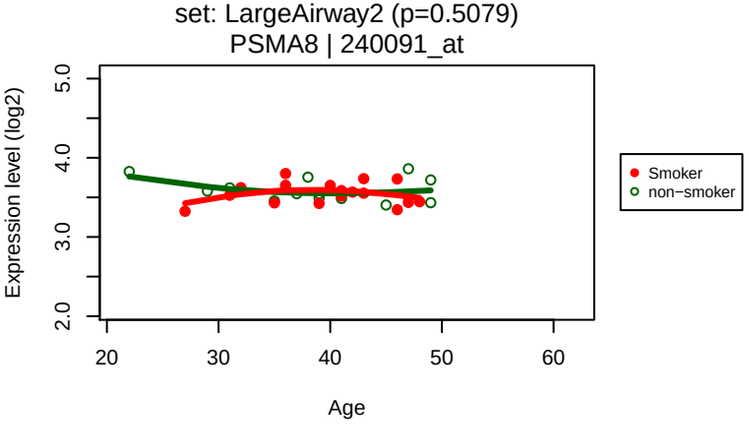

Gene expression ratio (smokers/nonsmokers)
gene: PSMA8 | 240091_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0601
p=0.0066
p=0.3123
p=0.2002
p=0.5079
p=0.0531
p=0.8855
p=0.1965
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

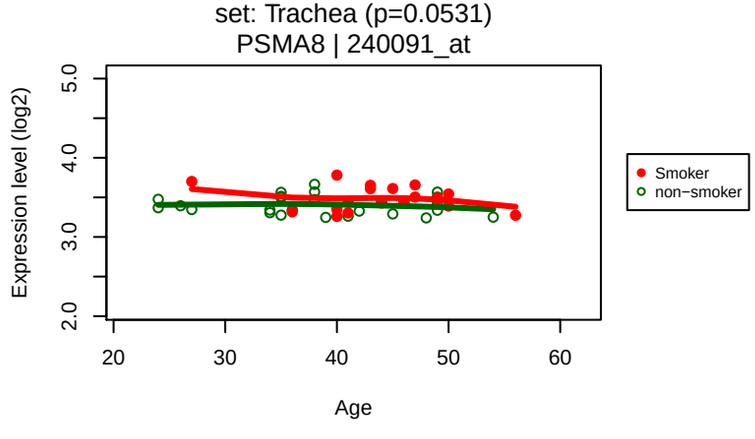

Gene expression ratio in Males
gene: PSMA8 | 240091_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0188
p=0.6521
p=0.2148
p=0.0151
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSMA8 | 240091_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3883
p=0.7341
p=0.4786
p=0.8386
SmallAirway2
LargeAirway2
Trachea
Macrophage2

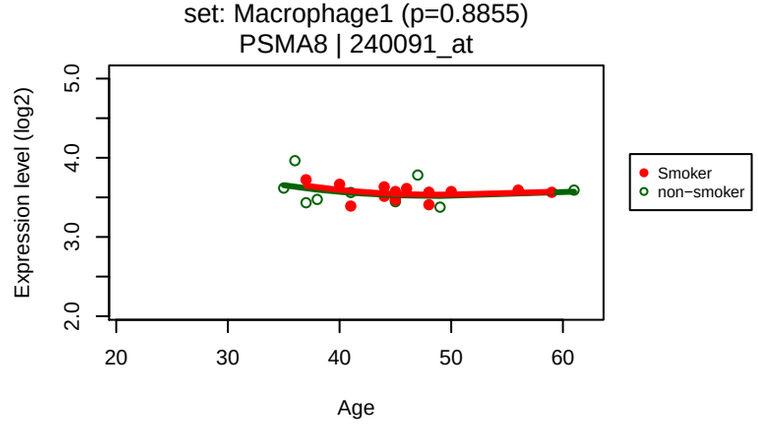

set: SmallAirway1 (p=0.0601)
PSMA8 | 240091_at
Expression level (log2)
Age
Smoker
non−smoker

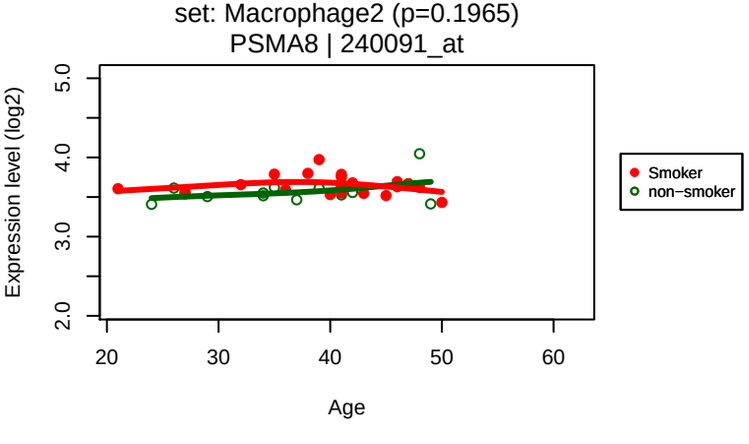

set: SmallAirway2 (p=0.0066)
PSMA8 | 240091_at
Expression level (log2)
Age
Smoker
non−smoker

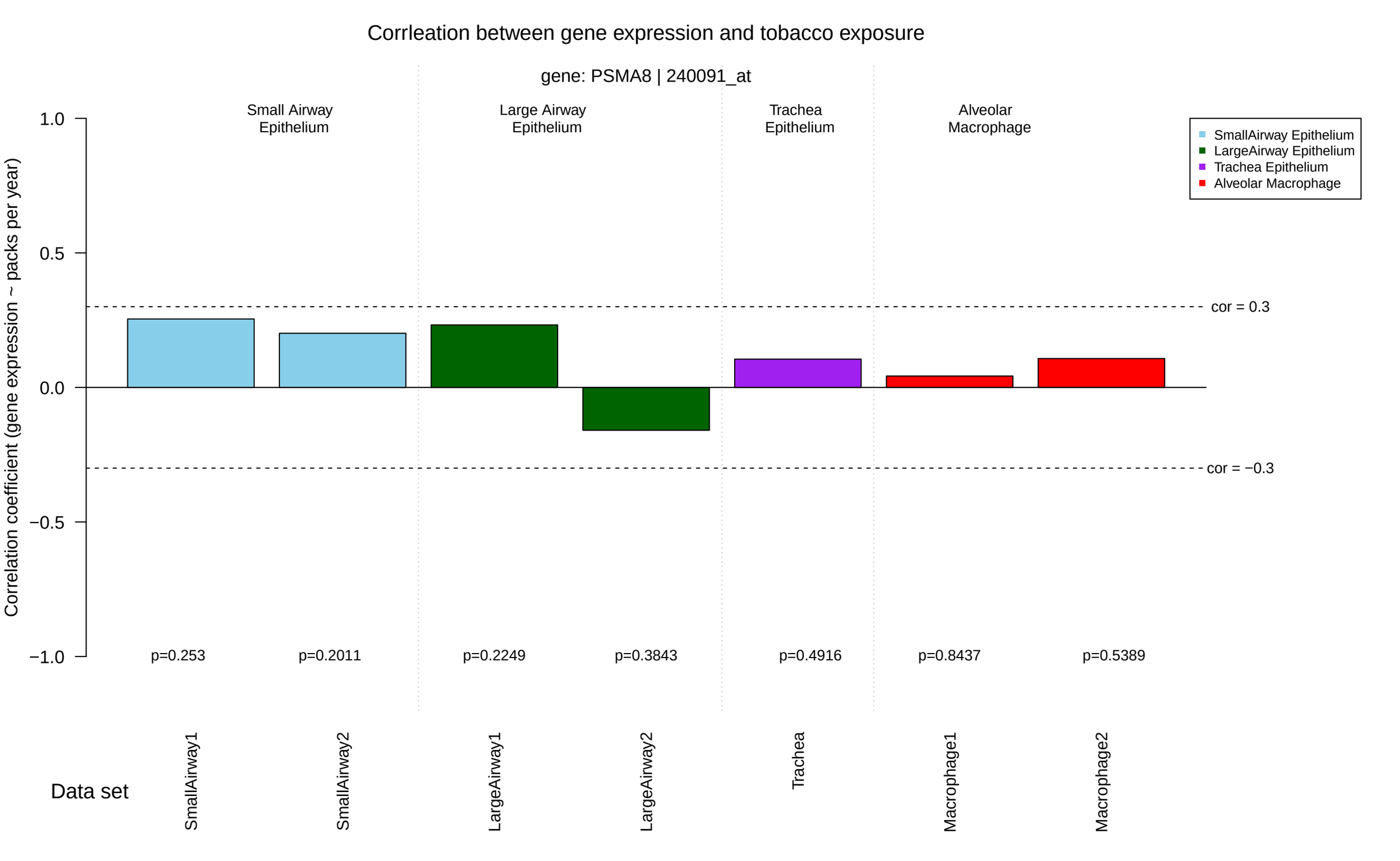

set: LargeAirway1 (p=0.2002)
PSMA8 | 240091_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5079)
PSMA8 | 240091_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0531)
PSMA8 | 240091_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8855)
PSMA8 | 240091_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1965)
PSMA8 | 240091_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSMA8 | 240091_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.253
p=0.2011
p=0.2249
p=0.3843
p=0.4916
p=0.8437
p=0.5389
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2